

Delta-CDR



SES 411

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Psyche-ESE Team

- Elysian Hudson - Team Leader
- Caleb Zick - Model Lead
- Daniel Joseph - Simulation Lead
- Jacob Coons - Computer Lead
- Spencer Emmons - Science Lead
- Syed Hamza Ayaz - Testing Lead



Agenda

- Science Question
- Background
- Motivation
- Science Traceability Matrix (STM)
- CONOPS
- System Block Diagram
- Requirement Flowdown
 - Mission-Level Requirements
 - Subsystem Definitions, Requirements, & Design
- Conclusion
- Next steps



Science Question



- ◆ By utilizing a computer simulation of a simplified model of 16-Psyche, determine whether it is feasible to map the subsurface via seismic wave propagation.



Background - 16 Psyche



- 16 Psyche is a large asteroid within the main asteroid belt, presumed to be largely composed of metal & hypothesized to be the partial core of a planetesimal.
- Named after the Greek goddess of the soul.
- An orbiter mission to investigate the asteroid is currently underway, scheduled to arrive in 2029.
- Current NASA Psyche mission:
 - Instrument Suite: Multispectral Imager, Gamma-ray and neutron spectrometer, Magnetometer, Gravity science investigation
- Mission precedents: NASA's InSight, Deep Impact, LCROSS missions. JAXA's Hayabusa2 mission.
- Single-site and/or multi-site seismology used to develop 3-dimensional maps of subsurface structures of planetary bodies.

Background - Software Package



- What is SeisSol?

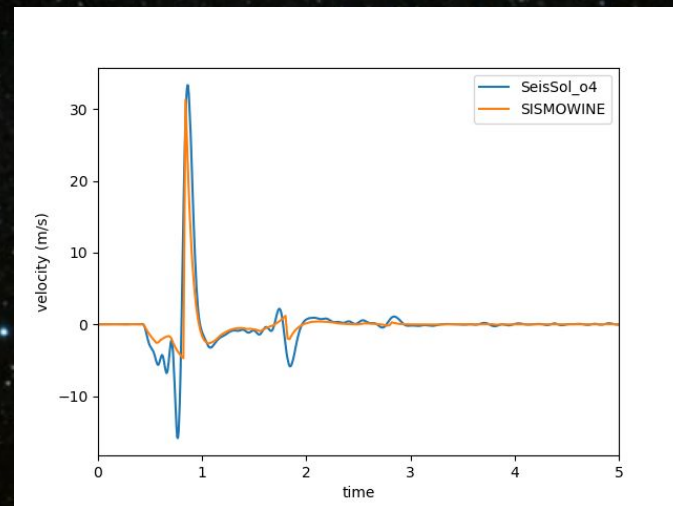
- A software package for simulating wave propagation based on the arbitrary high-order accurate derivative discontinuous Galerkin method (ADER-DG)

- What does it do?

- Utilizes meshes to approximate 3D model geometries
- Specifies material elasticity to approximate geological subsurface properties
- Produces reliable and accurate synthetic seismograms and other seismological data sets approaching standardized seismic benchmarks.

- Why does it work for us?

- Can select source wave generation point(s) with whatever energy we decide
- Can get detailed seismological data at any selected detection point (ground displacement and wave energy) in order to answer our science question.



SeisSol Documentation - WP2_LOH1

Motivation



- Very little is known about the interior structures of small planetary bodies.
- Knowledge about 16 Psyche currently relies on remote sensing observations.
- Directly investigate 16 Psyche to help determine if it is the core of a planetesimal, using geophysics and seismology to explore the subsurface.
- Possibly differentiated/layered.
- A highly sensitive seismometer could help constrain the number of small asteroids/meteoroids in the asteroid belt
- Learn more about the planetary formation/evolution.
- Future implications for planetary defense and resource exploitation.

Motivation–Decadal Survey



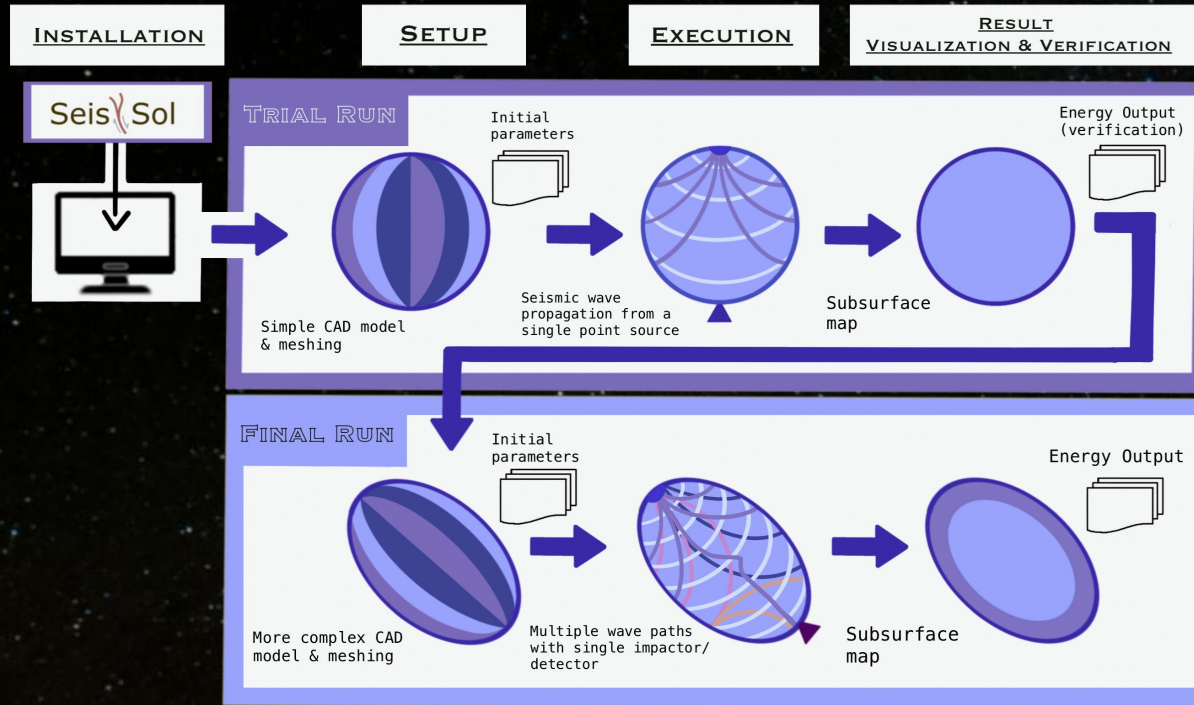
- 2023-2032 Decadal Survey Key Questions:
 - Q1 What were the initial conditions in the solar system, what processes led to the production of planetary building blocks, and what was the nature and evolution of these materials?
 - 16 Psyche is a presumed core, and therefore possibly a remnant planetary building block. Learning about its subsurface structure can tell us about the processes that lead to its formation, including processes such as differentiation.
 - Q4 How have the population of solar system bodies changed through time, how has bombardment varied across the solar system, and how have collisions affected the evolution of planetary bodies?
 - 16 Psyche may have experienced a hit-and-run event(s) during formation, and the subsurface structures can tell us about the impact history of the asteroid.
 - Q5 How did the interiors of solid bodies evolve, how is this evolution recorded in a body's physical and chemical properties, and how are solid surfaces shaped by internal and external processes?
 - 16 Psyche can be directly investigated to learn about the internal evolution of other solid bodies in our Solar System. Its internal structure will have recorded the processes responsible for its shaping.

Science Traceability Matrix (STM)

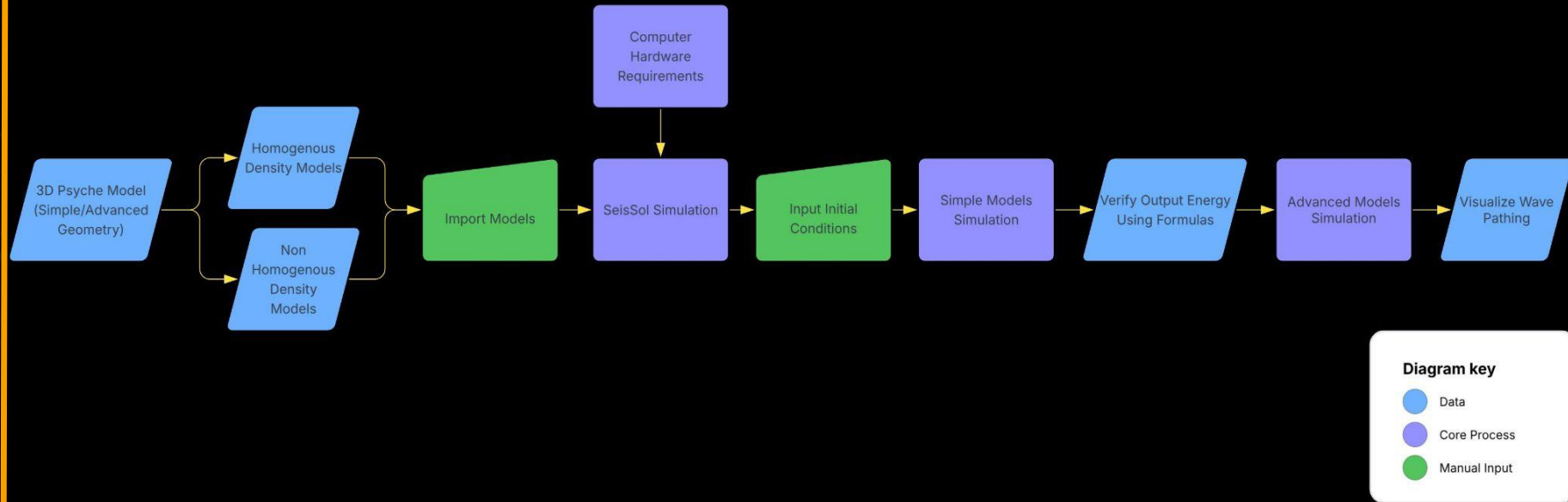


Objective	Measurement	Measurement Requirements	Technologies	Technological Requirements	Data Product
Determine if simulated seismic waves generated from an artificial impactor can be recorded by a virtual seismometer after traveling through 16 Psyche's interior	Simulated seismic waves that fully travel through the interior of the model	Calculated minimum kinetic energy of the impact is 4 MJ	-Software -CPU -Memory -Storage -CAD Model	<u>Software</u> -SeisSol Simulation Software and dependencies (see software subsystem slide)	Seismic wave energies at selected measurement point
	Detect simulated seismic wave amplitude at selected point on the model of 16 Psyche	Minimum measurable ground acceleration spectral density of: $5 \times 10^{-9} \text{ ms}^{-2} \text{ Hz}^{-1/2}$		<u>CPU</u> -Single chip PC Setup -Lenovo: Intel Core i7-7600 -PC: TBD <u>Memory</u> -Lenovo: 16GB RAM for test model -PC: TBD for final model <u>Storage (PC)</u> -2TB SSD: Operation -4TB HDD: Long-term Storage	Surface displacement at selected measurement point
	Propagation of seismic waves	Maximum deviation from expected wave pathing of no more than 1 degree		<u>CAD Model</u> -Minimum accuracy of bilayer, non-homogeneous, order of magnitude density estimate of 16 Psyche	Animated visualization of seismic waves traversing the model (.xdmf)

Psyche-ESE CONOPS



System Block Diagram



Requirements Levels

L1: Mission and Measurement Requirements

L2: Subsystem Requirements



Requirements

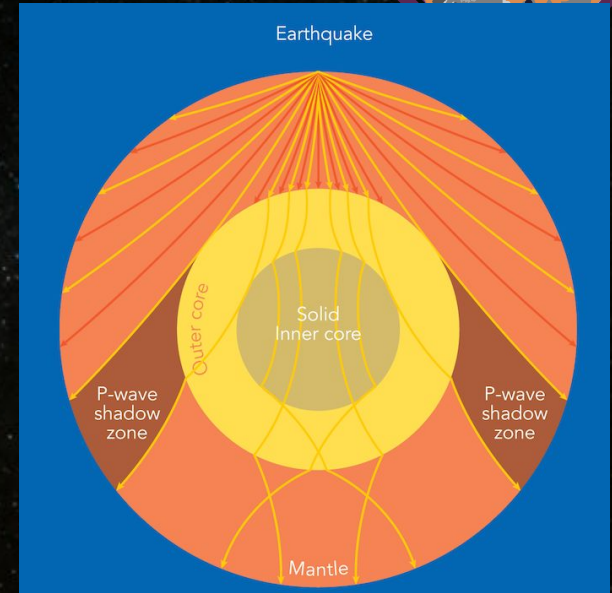


Mission Requirements - Level 1:

ID	Requirement	Verification	Parent
L1.1	Propagation of seismic waves - The simulation shall model, to within 10% of expected parameters, the propagation of seismic waves through a 3d body.	Test, Analysis, Demonstration	Science Question
L1.2	Computer Specifications - The computer used to run the simulation(s) shall be able to run the simulation of a minimum 16 Psyche model complexity within 1 day (24 hours)	Test, Demonstration	Science Question
L1.3	Initial Conditions - The simulation code shall be able to support varying initial conditions within the simulated instruments and simulated asteroid.	Inspection	Science Question

L1 Requirement – 1.1

- Propagation of seismic waves - The simulation shall model the propagation of seismic waves through a 3D body.
- Shall be tested using a homogeneous sphere and the results compared to manually calculated results.
- In order to be considered accurate, the model should agree with the calculated results to within 10%.
- Simulation takes into account:
 - Wave attenuation
 - Reflections/refractions at boundaries



University of Hawaii. (n.d.). *Compare-Contrast-Connect: Seismic Waves and Determining Earth's Structure*. Exploring Our Fluid Earth.

L1 Requirement – 1.2

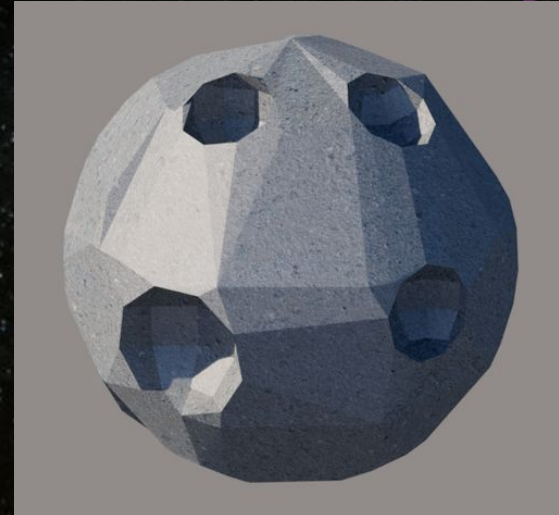
- Computer Specifications - The computer used to run the simulation(s) shall be able to run the simulation of a minimum 16 Psyche model complexity within 1 day (24 hours)
- Considerations:
 - Hardware (accessibility, performance, compatibility with software)
 - Software (CPU architecture, package management, Operating System, file I/O)



PC

L1 Requirement – 1.3

- Initial Conditions - The simulation code shall be able to support varying initial conditions within the simulated instruments (InSight SEIS) and simulated asteroid.
- Includes:
 - Adaptable Model Characteristics:
 - Density
 - Layering
 - Varying Inputs:
 - Impactor Characteristics
 - Kinetic Energy
 - Time and Wave Frequency
 - Location



Romanenko, K. (2019, June 17). *asteroid-5*.
GrabCAD.

Subsystem Requirements - 1.1

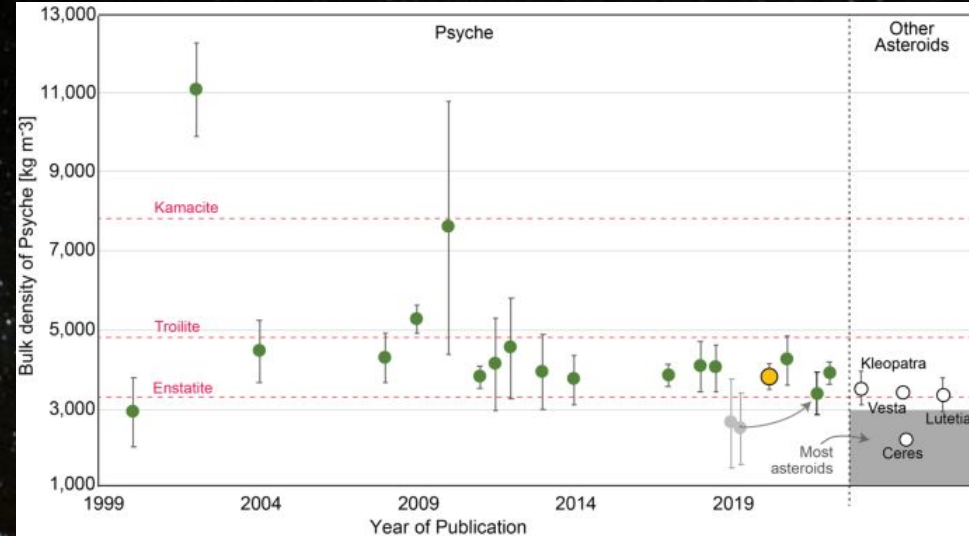


ID	Requirement	Verification	Parent	Rationale
L2-WAVE-1	Basic tests of the simulation code shall use homogeneous spheres of various densities approximating 16 Psyche - a radius of 115 km	Test, Analysis	L1.1	Confirms the accuracy of the simulation code
L2-WAVE-2	The readings at the simulated detector shall be within the sensitivity limits of the SEIS instrument, with a ground displacement of $1 \text{ \AA}$ and a ground acceleration spectral density of $5 \times 10^{-9} \text{ m s}^{-2} \text{ Hz}^{-1/2}$	Demonstration, Analysis	L1.1	Confirms the waves could be detected by a real-world seismometer

Science Subsystem - Physical Parameters



- Gravitational Acceleration
 - Mean: 0.144 m/s^2
 - Min: 0.106 m/s^2 , max: 0.145 m/s^2 (Shepard et al. 2017)
- Asteroid Size
 - Longest: 279 km
 - Shortest: 189 km (Shepard et al. 2017)
- Main Variable:
 - Composition/Density
 - Min: 2.5 g/cm^3
 - Mid: 3.9 g/cm^3
 - Bulk Density $3.4\text{-}4.4 \text{ g/cm}^3$ (Elkins-Tanton et al. 2020)
 - Max: 8 g/cm^3



Elkins-Tanton, Linda T., et al. "Distinguishing the Origin of Asteroid (16) Psyche - Space Science Reviews." *SpringerLink*, Springer Netherlands, 12 Apr. 2022

Science Subsystem - Physical Parameters Cont.



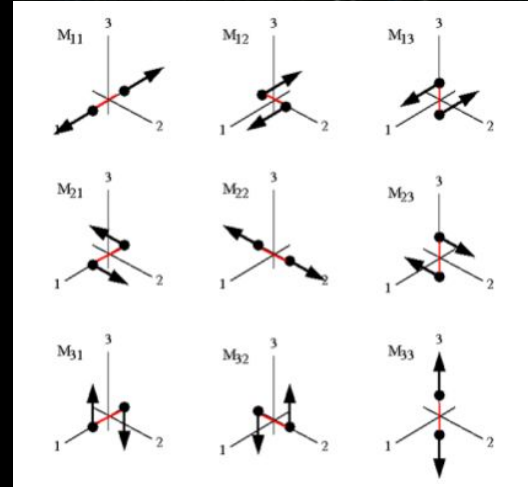
- Source Time
 - Depends on characteristics of wave source
 - We will use: .01 s
- Frequency
 - Range:
 - Low: 0.01 Hz
 - High: 100 Hz
 - Central
 - 1 Hz
 - Ratio
 - 10000
- Impact Kinetic Energy
 - At least 4 MJ

$$M = \sqrt{\frac{2}{3}} \begin{bmatrix} M_0 & 0 & 0 \\ 0 & M_0 & 0 \\ 0 & 0 & M_0 \end{bmatrix}$$

- Seismic Moment Tensor
 - (Nishiyama et al. 2020)

$$M_0 = \left(\frac{k E_I}{4.5 \times 10^{-9}} \right)^{\frac{1}{1.24}}$$

- k = seismic efficiency
 - We assume 10^{-5}
- E_I = impact kinetic energy

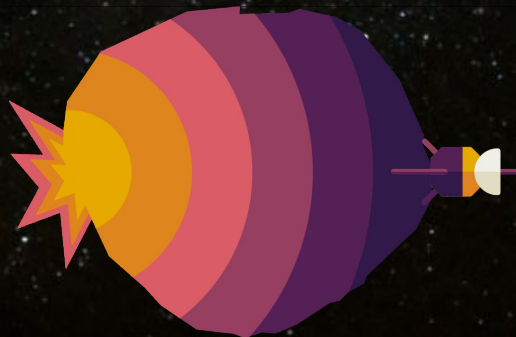


"Moment Tensors - A Practical Guide: MXrap."
 mXrap RSS, Australian Center for Geomechanics,
 10 Nov. 2025

Science Subsystem - Question Answers

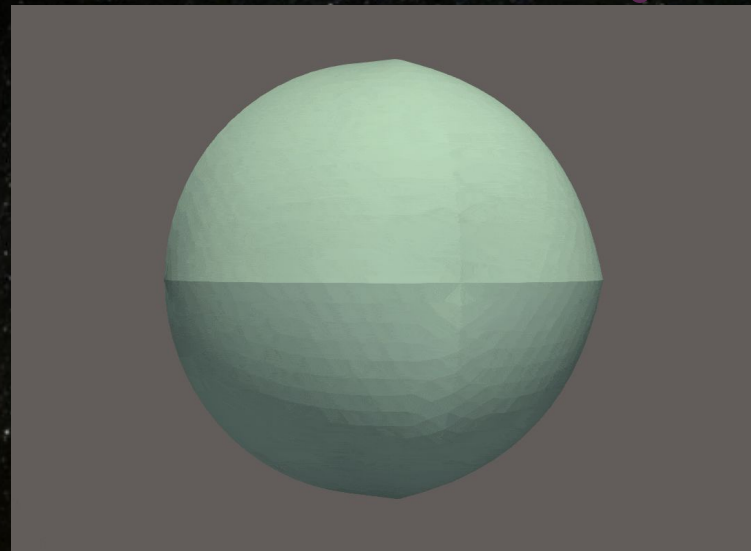


- Data Visualization
 - ParaView
- Signal strength
 - What is the ground acceleration and frequency at the location of the detector
- Comparison with noise floor
 - If the ground acceleration is equal to or greater than the minimum detection threshold we have set (based on the SEIS) the science question will be answered “yes”
 - If it is less, then the answer will be “no”



Model Subsystem

- Beyond the parameter files, SeisSol needs two things to begin a simulation:
 - A 3D mesh environment, which can be created using Gmsh.
 - PUMML and xdmf files, both created from the original mesh.
- Using Gmsh, a mesh can be created from 3D geometry using tetrahedral elements.
 - Final number of mesh elements is still TBD
 - Preliminary testing has gone up to 30,000 elements

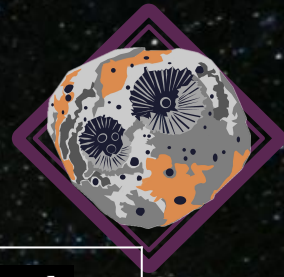


Model Subsystem Interfaces

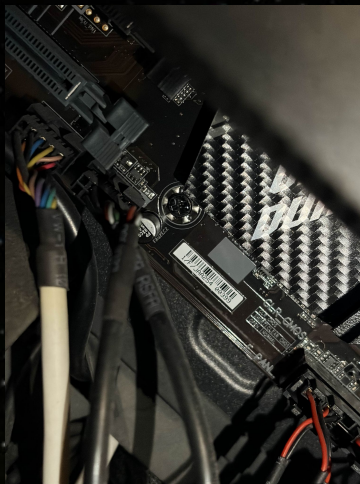


- Input Interfaces
 - SeisSol Software
 - Dependencies
 - Gmsh Software
 - Dependencies
 - Defines shape of model, size, layering, etc.
 - PUMGen Software
 - Dependencies
 - Converts mesh into PUMML format
- Output Interfaces
 - 3D Geometry Model (.geo)
 - Internal Surface File (.msh4)
 - SeisSol-PUMML Files (.puml.h5/.xdmf)

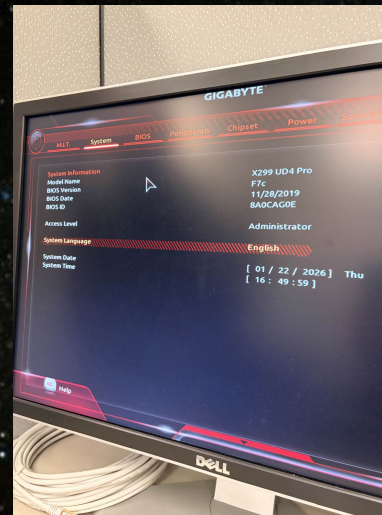
Subsystem Requirements - 1.2



ID	Requirement	Verification	Parent	Rationale
L2-COMP-1	The OS and other dependency software shall support the successful operation of SeisSol	Test	L1.2	Prevents compilation and runtime errors
L2-COMP-2	The hardware models and setup shall meet the minimum performance required to successfully complete a run of SeisSol lasting no longer than 24 hours	Test	L1.2	Allows for multiple runs of various 16 Psyche Models



Disclaimer...



Computer Subsystem - Software



- SeisSol - Our primary seismic wave simulation software.
- Operating System - SeisSol was developed on a unix-based OS. For maximum compatibility, we should run it on the same OS.
- Package Management - SeisSol utilizes Spack Open Source Package Manager for installation of the program, as well as all of its dependencies.
- CPU Architecture - Host architecture and corresponding command is needed for compiler tuning variables and utilization of proper code generators upon build phase, allowing for optimal build performance. See software key components for the current architecture used to build on the Lenovo.

Computer Subsystem - Hardware



- CPU - Performance reliant on:
 - **Number of CPUs in Setup:** We have chosen a single CPU setup in both the Lenovo and the PC tower.
 - **Specifications**
 - **Lenovo:** Intel Core i7-7600u
 - **PC:** TBD
- Memory (RAM) - Performance reliant on:
 - **Number of Gigabytes** - There is a minimum amount of RAM needed to meet our runtime demands
 - **Lenovo:** 16GB RAM for test model
 - **PC:** TBD

Computer Interfaces



- Input Interfaces
 - Environment
 - Temperature control
 - Minimal contaminants (dust, moisture)
 - Stable and mount or stand to ensure no interruptions to or damaging of hardware.
 - Power draw from lab (operation, and performance)
 - Keyboard, mouse - interact with OS via user interface (UI)
 - UI - interact with and instruct software to perform operations and run simulation
 - Remote Access Software
- Output Interfaces
 - USB, USB-C, DP, HDMI or some other streaming methodology that properly displays the UI on a screen and, by extension, the results of the simulation.
 - Remote Access Software

CPU (Lenovo)

- Intel Core i7-7600u, laptop model
 - Thermals may be suboptimal for performance in this casing
- 2 cores
- 4 threads
- Base Frequency - 2.80 GHz
- Boost Frequency - 3.90 GHz
- Max Supported Memory - 32 GB
- Supported Memory Bandwidth - 34.1 GB/s
- Supported Memory Types (for maximum bandwidth above) - DDR4-2133, LPDDR3-1866, or DDR3L-1600



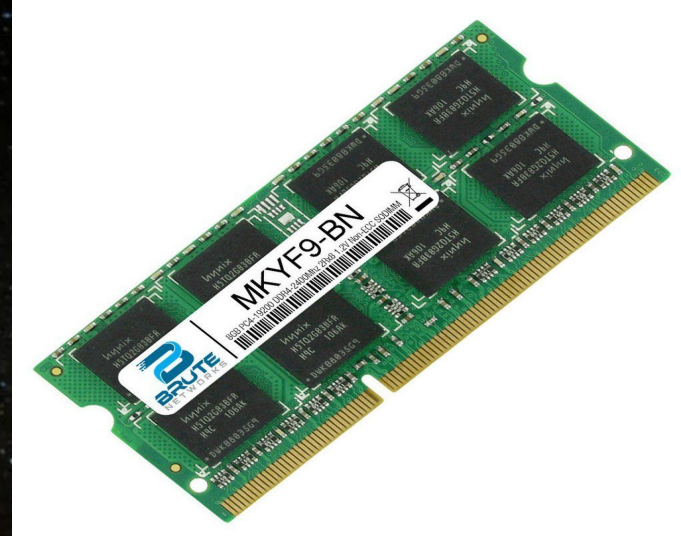
CPU (PC)

- TBD



RAM (Lenovo)

- 2, 8GB RAM Sticks (part #: MKYF9)
- 16GB Total of RAM
- DDR4 - second best double data rate generation
- Clock speed (frequency) - 2400 MHz
- Current test runs show more RAM is needed (sim is exiting due to running out)
- Could upgrade system to 32 GB of RAM (slots and budget permitting)



RAM (PC)

- TBD



Storage (PC)



- Simulation and System Operation

- Spec: Samsung 990 PRO M.2 2280 NVMe SSD
- Size: 2TB
- Read: 7,450 MB/s
- Write: 6,900 MB/s



- Long-term Data Storage

- Spec: Seagate IronWolf NAS HDD (3.5", CMR)
- Size: 4TB
- Read: 180 MB/s
- Write: 202 MB/s



Software Key Components



- OS
 - Any recent, stable Linux distribution will work.
 - Current distribution on Lenovo - Ubuntu 22.04.5 LTS
- Package Management
 - Linux has built in package managers
 - Official recommendation in documentation for SeisSol is SPACK
- CPU architecture Lenovo
 - Lenovo
 - The Intel Core i7-7600u is based on the Kaby Lake architecture
 - SeisSol does not support Kaby Lake specifically, but does support other i7 architectures
 - Current SeisSol build was compiled with the Intel Haswell architecture
 - “HOST_ARCH = hsw”
 - PC
 - TBD

Computer Testing



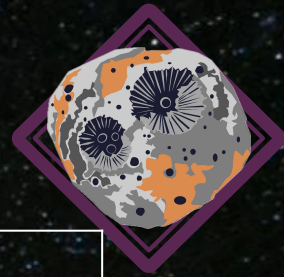
- **Test COMP-01: SeisSol Operational Test**

- Purpose: Ensure that SeisSol was properly installed and compiled
- Test: Run the following command line prompt, directed at the proper installation folder:
 - `~/SeisSol/build-release/SeisSol_Release_dhsw_6_elastic --help`
- Pass/Fail
 - Pass: Return message consisting of a welcome message, copyright, version, build date, last commit ID, and some other qualifiers
 - Fail: No terminal reply, a “no such file or directory” bash message, or any other error message

- **Test COMP-02: Hardware Performance Test**

- Purpose: Ensure that the runtime does not exceed 24 hours for the minimum model complexity that is acceptable in answering our science question
- Test: Iterative. Start with a test model (100-200 mesh elements), then increase complexity in subsequent runs up to the accepted level of complexity. Keep track of runtime for each iteration.
- Pass/Fail
 - Pass: All iterations exit runs in 24 hours or less
 - Fail: Any of the iterations take longer than 24 hours

Subsystem Requirements - 1.3



ID	Requirement	Verification	Parent	Rationale
L2-SIM-1	The simulation code shall be able to support multiple receivers and varying positions	Inspection	L1.3	Shows effect of varying receivers on data
L2-SIM-2	The simulation code shall be able to support varying impact angles, positions, and energies.	Inspection	L1.3	Shows effect of varying impacts on data
L2-SIM-3	The simulation code shall be able to support varying density, layering, impact, energy efficiency, and porosity of 16 Psyche models	Inspection	L1.3	Shows effect of varying conditions of 16 Psyche on data

Simulation Subsystem



- Custom CAD importing and meshing
- Simulate point source impact
- Allow for control over initial conditions (model density/layering, impact source point and detector position, initial energy, impact angle)
- Visualize wave pathing through 3D Psyche model
- Output total energy of simulation

Simulation Subsystem Interfaces



- Input Interfaces
 - Initial Conditions Parameter File
 - Model density and layering
 - Impact energy
 - Seismic source type
 - Model Mesh
 - Input mesh map for Psyche model
- Output Interfaces
 - Wavefield data
 - Visualization metadata files for use in Paraview

Simulation Subsystem Testing



- **Test SIM-01: SeisSol Initial Conditions**
 - Purpose: Determine level of control for desired initial conditions
 - Test: Write an “easi” file script using the “easi” file directory to control desired initial conditions and mesh
 - Pass: “easi” parameter file scripted to control number of seismic receivers on our mapped model and their positions; point-source seismic origin; model density, layering, seismic magnitude, energy efficiency, and model porosity
- **Test SIM-02: Simple Geometry/Mesh**
 - Purpose: Use imported spherical model of 16 Psyche and apply different meshes and conditions to run simulations
 - Test: Open a terminal, `cd ~/my_simulationSimple`, then:
`mpirun-np1~/SeisSol/build-release/SeisSol_Release_dhsw_6_elastic parameters.par`
 - Pass: Saved wavefield output file and surface output metadata files for visualization in Paraview, detectable readings

Simulation Subsystem Testing Cont.



- **Test SIM-03: Complex Geometry/Mesh**
 - Purpose: Use imported similar model of 16 Psyche and apply different meshes and conditions to run simulations
 - Test: Open a terminal, `cd ~/my_simulationComplex`, then:
`mpirun-np1~/SeisSol/build-release/SeisSol_Release_dhsw_6_elastic parameters.par`
 - Pass: Saved wavefield output file and surface output metadata files for visualization in Paraview, detectable readings

Testing Subsystem



- To confirm our results, we can compare the the energy from our test simulation with our calculations.
- The Seismic wave energies can be calculated by using the initial values by hand
- By using the values used for:
 - v_p = longitudinal wave velocity
 - v_s = transverse wave velocity
 - F = momentum of impactor
 - τ = time constant
 - ρ = density
- The total Seismic Wave Energy ($v_s + v_p$) equation can be separated into energies for the longitudinal and transverse waves

$$E_s(t) = \frac{F^2 \pi^{3/2}}{2^{5/2} \tau^3 \rho} \left(\frac{1}{3v_p^3} + \frac{2}{3v_s^3} \right),$$

Courville, S. W., & Sava, P. C. (2020). Speckle noise attenuation in orbital laser vibrometer seismology. *Acta Astronautica*, 172.

Testing Subsystem Interfaces



Only Simulation data and calculations, no external input, use same initial values as the simulation

Inputs:

- Longitudinal seismic-wave speed (v_p)
- Shear Wave speed (v_s)
- Density
- Impactor momentum
- Time constant
- Model layers and geometry
- Impactor and detector locations and distance
- simulated detector energy + arrival times.

Outputs:

- ideal energy calculation,
- % difference between simulated and calculated values
- pass/fail based on requirements.

Testing Plan



Test 1 - Energy Accuracy

- Compare simulation energy with calculated ideal value.
- Must be within 10%.
- If not, the run fails.

Test 2 - Minimum ground acceleration detected

- Check if ground at detector is accelerated to at least $5 \times 10^{-9} \text{ m/s}^2$.
- If not, the run fails.

Test 3 - Wave Timing

- Compare simulated wave arrival time (for v_s and v_p) with expected travel time.
- Time should be close to the calculated value (10%).

List of Tracked Risks



ID	Summary	L	C	Trend	Approach	Risk Statement
1	Power	1	3	↓	R	The system loses power during runtime
2	Budget	1	5	↓	W	The project exceeds a budget of \$2000
3	Hardware	2	4	↓	M	Computer suffers major component failure
4	Schedule	2	5	↓	M	Unexpected delays (installation, user error)
5	Software	1	3	↓	R	The software suffers a runtime failure or corruption
6	Facilities	1	3	↓	R	Our group cannot get access to facilities to base out of
7	Access	1	4	↓	W	Can't get access to higher performance hardware if needed

L = Likelihood (1-5)

1 = not likely

5 = extremely likely

C = Consequence (1-5)

1 = low consequence

5 = high consequence

LxC Trend

↓ - Decreasing (improving)

↑ - increasing (worsening)

→ - unchanged

NEW - added this week

Approach

A - accept

M- mitigate

W - watch

R - research

Risk Matrix



Criticality
HIGH
MED
LOW

L I K E L I H O O D	5					
	4					
	3					
	2			3	4	
	1		1, 5	7	2	
		1	2	3	4	5
CONSEQUENCES						

Schedule



TASK	START	END
Computer & Installation		
Access & inspect computer	12-Jan-26	26-Jan-26
Find lab space	12-Jan-26	26-Jan-26
Purchase & install parts	26-Jan-26	16-Feb-26
Install SeisSol	16-Feb-26	23-Feb-26
Simple Modelling & Meshing		
Build simple CAD model & mesh	12-Jan-26	26-Jan-26
Advanced Modelling & Meshing		
Build advanced CAD model & mesh	26-Jan-26	23-Feb-26
Trial Run & Verification		
Debugging & running	23-Feb-26	9-Mar-26
Verifying output results	9-Mar-26	23-Mar-26
Final Run & Results		
Debugging & running	23-Mar-26	6-Apr-26
Output results	6-Apr-26	20-Apr-26



Budget



- Arizona State University offer access to computers (including the Sol supercomputer) for free for all students.
- Current selections for simulations are open source aside from backup plan.

Category	Item	Cost	Qty	Total
<i>Electronics</i>	2TB Samsung 990 PRO M.2 2280 NVMe SSD	\$299.99	1	\$299.99
	4TB NAS Internal Hard Drive	\$99.99	1	\$99.99
<i>Cleaning/Maintenance</i>	Canned Air	\$16.99	1	\$16.99
	Anti-Static Gloves (12-Pack)	\$17.50	1	\$17.50
	Microfiber Towels	\$8.49	1	\$8.49
	All Purpose Cleaner	\$3.97	1	\$3.97

TOTAL COST	\$446.93
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Conclusions & Next Steps



- Our science question depends on a simulation of probable internal conditions and structural layers of 16-Psyche, and necessitate the use of SeisSol to produce a reasonable map of 16-Psyche's interior.
- Continue researching backup plans (InstaSeis & SeismoSignal 3D)
- Next steps:
 - Purchase and obtain parts (budget/parts list already sent)
 - Replace current hard drive/SSD in desktop
 - Begin installation of remote access & SeisSol on desktop
 - Contact SeisSol developers

**Thank
You!
Questions?**



Technical Readiness Level (?)



Testing Key Components



Energy Calculation:

- Manually calculate ideal energy using equation.
- Use same initial values as the simulation

Result comparison:

- Calculate the value difference between ideal and simulated energy.
- Must be within 10% to count as accurate.

Minimum Detection Threshold:

- Detector must accelerate by $5 \times 10^{-9} \text{ m/s}^2$

Wave Behavior

- Compare the timing and speed of waves (v_s and v_p) in simulation to the calculated expectation
- Should be within 10% to be verified.

Backup Slides:



Requirements Must be SMART:

- Specific -
 - It must address only one aspect of the system design or performance
 - It must be expressed in terms of the need (what and how well), not the solution (how).
- Measurable -
 - Performance is expressed objectively and quantitatively
 - For example, an exact pointing requirement (in degrees) can be tested thus verified prior to launch.
- Achievable -
 - It must be technically achievable at costs considered affordable
 - Note: JWST early designs specified an aperture requirement eventually descope due to technical issues with deployment.
- Relevant -
 - It must be appropriate for the level being specified
 - For example, requirement on the solar cells should not be designated at the spacecraft level.
- Traceable -
 - Lower level requirements (children) must clearly flow from and support higher level requirements (parents).
 - Requirements without a parent are referred to as orphans, and need to be assessed for necessity of inclusion.

STM Contents

(Groppi)

- Teams developing NASA missions use an STM to summarize the science and top-level requirements.
- An STM has the following columns:
 - **Science Objectives:** The elements of your science question(s)
 - **Measurement Objectives:** What quantities do you need to measure to achieve the science objective:
 - **Science Measurement Requirements:** What do you need to measure and how well do you need to measure it? Expected measurement range and required measurement error.
 - **Instrument Functional Requirements:** What other requirements does the instrument that takes the data need to have to meet your needs?
 - **System Functional Requirements:** What other requirements does the whole experiment need to have to meet your requirements?
 - **Science Product:** What is the data your experiment will produce (the low-level data measured, not the answer you get after you analyze it).

Basic simulation

It consists of:

- 1 Impact point
- 1 detection point
- 2 -3 layers

An example of SeisSol simulation for 2 layers

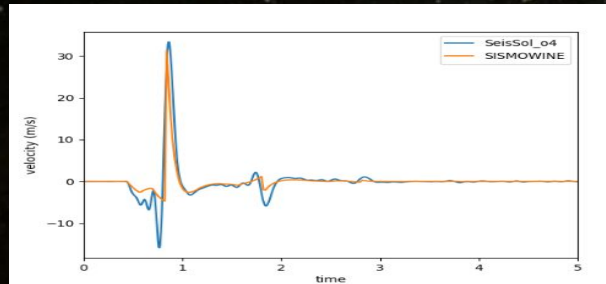
From central point (0,0,0)

Layer: from 0 to 1000 (z direction);

Half-space: 1000 to 3300 (z direction)

Impact source: (0, 0, 2000).

We can pick any impact and detection point by specifying the coordinates,

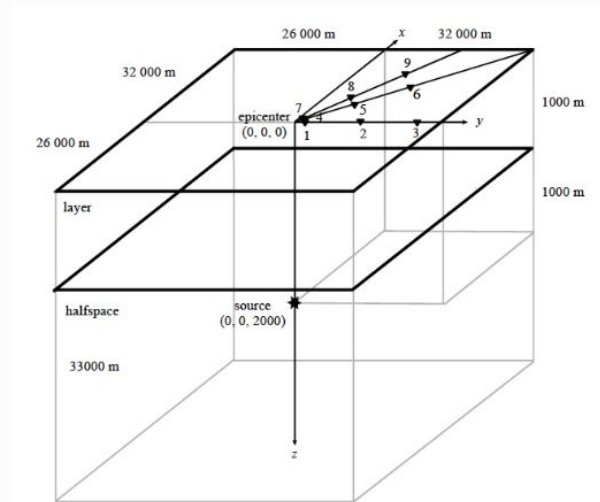


Example of a basic simulation of LOH1



	Vp (m/s)	Vs(m/s)	density	Qp	Qs
layer	4000	2000	2600	Inf	Inf
half-space	6000	3464	2700	Inf	Inf

Table: Material properties in LOH1.



Geometry of LOH1.

Requirements



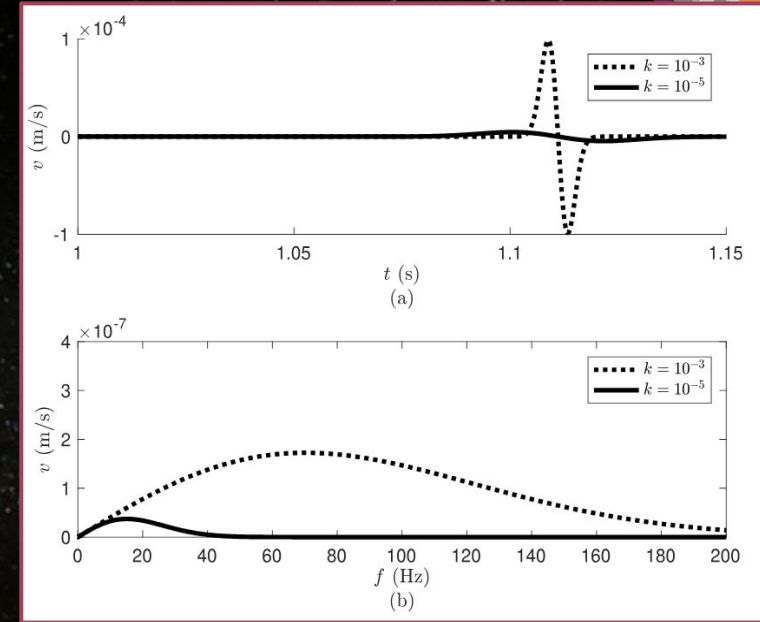
Mission Requirements - Level 1:

ID	Requirement	Verification	Parent
MR1.1	Propagation of seismic waves - The simulation shall to accurately model the propagation of seismic waves through an artificial 3d body.	Output of test model matches manually calculated tests to 1 Order of Magnitude	Science Question
MR1.2	Computer Specifications - The computer used to run the simulation(s) shall be able to run the simulation of a minimum 16 Psyche model complexity within 1 day (24 hours)	The simulation completes 1 run in 24 hours or less	Science Question
MR1.3	Initial Conditions - The simulation code must be able to support varying initial conditions within the simulated instruments and simulated asteroid.	The simulation code can be run repeatedly with varying initial conditions	Science Question

Mission Requirement – 1.1



- Use previously acquired data to estimate frequency range, needed sampling rate, and timing accuracy.
- Use finite element models and seismic wave propagation simulations of Psyche's interior to refine previously estimated data.
- Use simulations to determine optimal seismometer placement.
- Measure changes in seismic signal characteristics as they propagate to map the subsurface interior.



Courville, S. W., & Sava, P. C. (2020). Speckle noise attenuation in orbital laser vibrometer seismology. *Acta Astronautica*, 172.

Mission Requirement – 1.2



- Impact device shall generate enough energy to propagate seismic signal throughout the entirety of 16 Psyche.
- Requires multiple preliminary assumptions that are not yet known to be accurate to 16 Psyche.
- Simulation will be needed to better estimate 16 Psyche's wave velocities, seismic efficiency of impact, $k = E_s / E_k$, and other key values.

$$E_s(t) = \frac{F^2 \pi^{3/2}}{2^{5/2} \tau^3 \rho} \left(\frac{1}{3v_p^3} + \frac{2}{3v_s^3} \right),$$

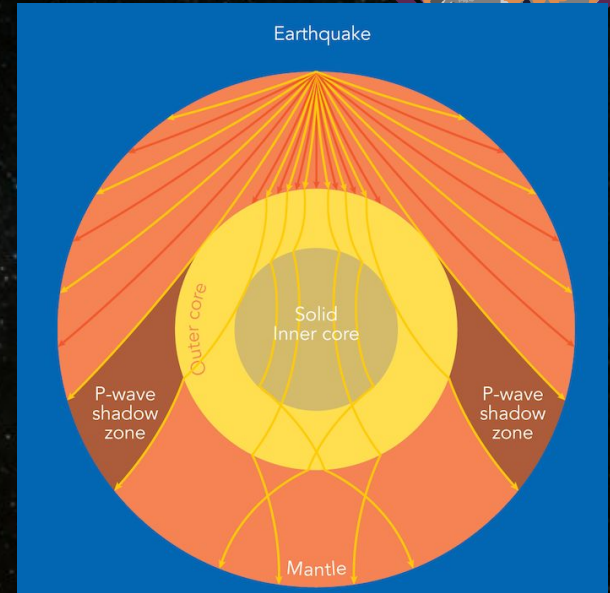
Where

- F = momentum of impactor
- v_p = longitudinal wave velocity
- v_s = transverse wave velocity
- τ = time constant
- ρ = density of target surface

Courville, S. W., & Sava, P. C. (2020). Speckle noise attenuation in orbital laser vibrometer seismology. *Acta Astronautica*, 172.

Mission Requirement – 1.3

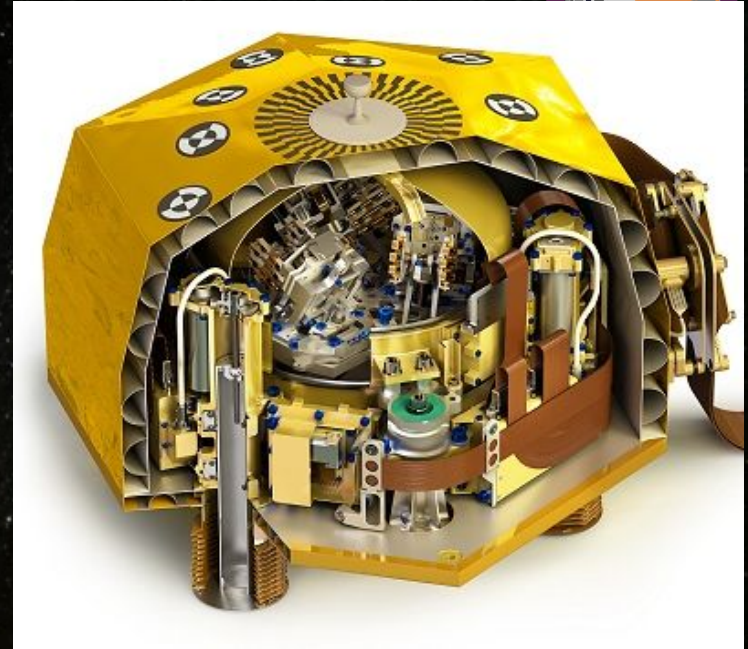
- The simulation shall be able to account for varying numbers and positions of impactors and/or detectors.
- More impactors/detectors will improve precision of measurements.
- But larger numbers will also be less practical.



University of Hawaii. (n.d.). *Compare-Contrast-Connect: Seismic Waves and Determining Earth's Structure*. Exploring Our Fluid Earth.

Science Subsystem

- Accuracy - The simulation code's accuracy will be confirmed by comparison to manually calculated results in simple examples
- Measurability - The simulated seismic waves at the location of the detector must be above the noise floor of real-world detectors
 - SEIS - The SEIS (Seismic Experiment for Internal Structure) is a seismometer on Mars included with the InSight Mission - due to its proven use on a non-Earth object it will be our real-world reference point
 - The SEIS is sensitive to a ground displacement of $1 \text{ \AA}$ and ground acceleration spectral density of $5 \times 10^{-9} \text{ ms}^{-2} \text{ Hz}^{-1/2}$



Interactive VBB pendulum. L'instrument seis - SEIS / Mars InSight. (2018, February 26).